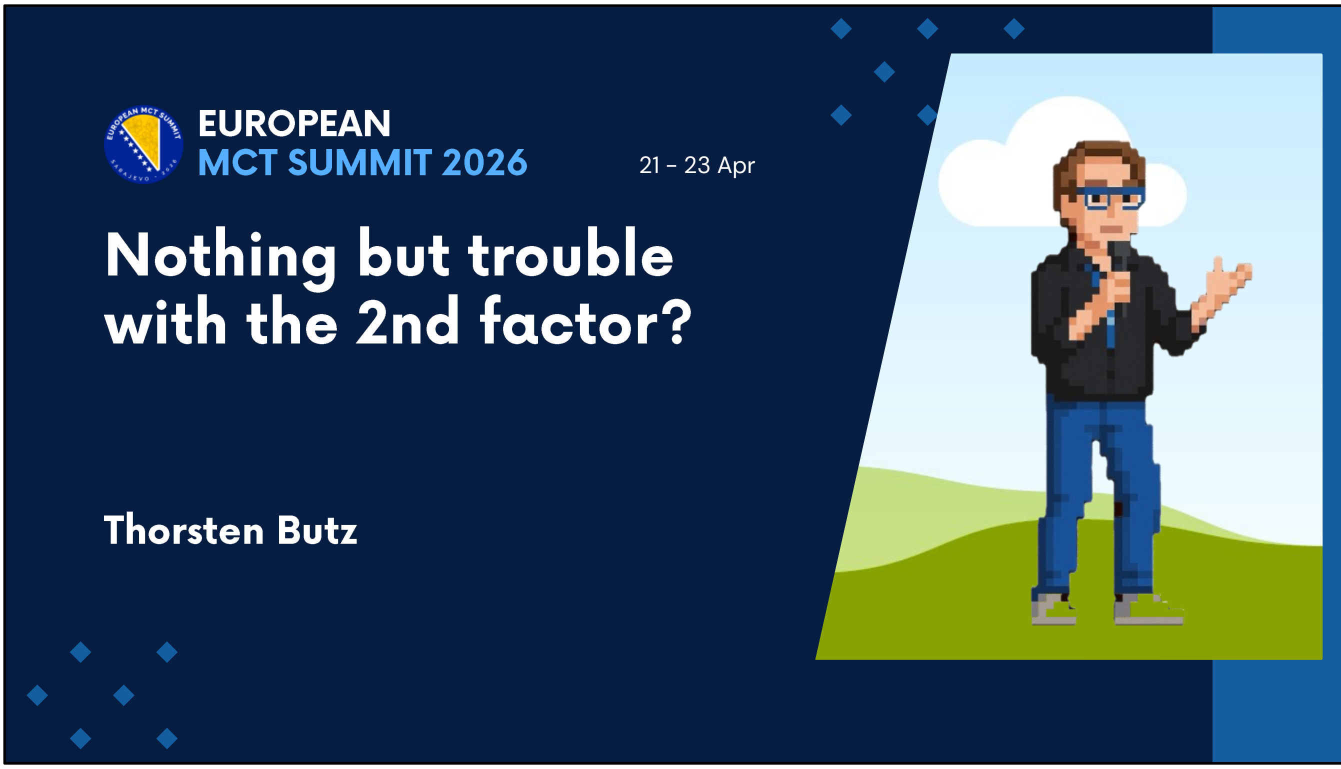

Nothing but trouble with the 2nd factor?

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From TOTP to passkeys and from Fido 2 tokens to QR codes, we explore simple tricks to help you and your students effortlessly connect to the online world.

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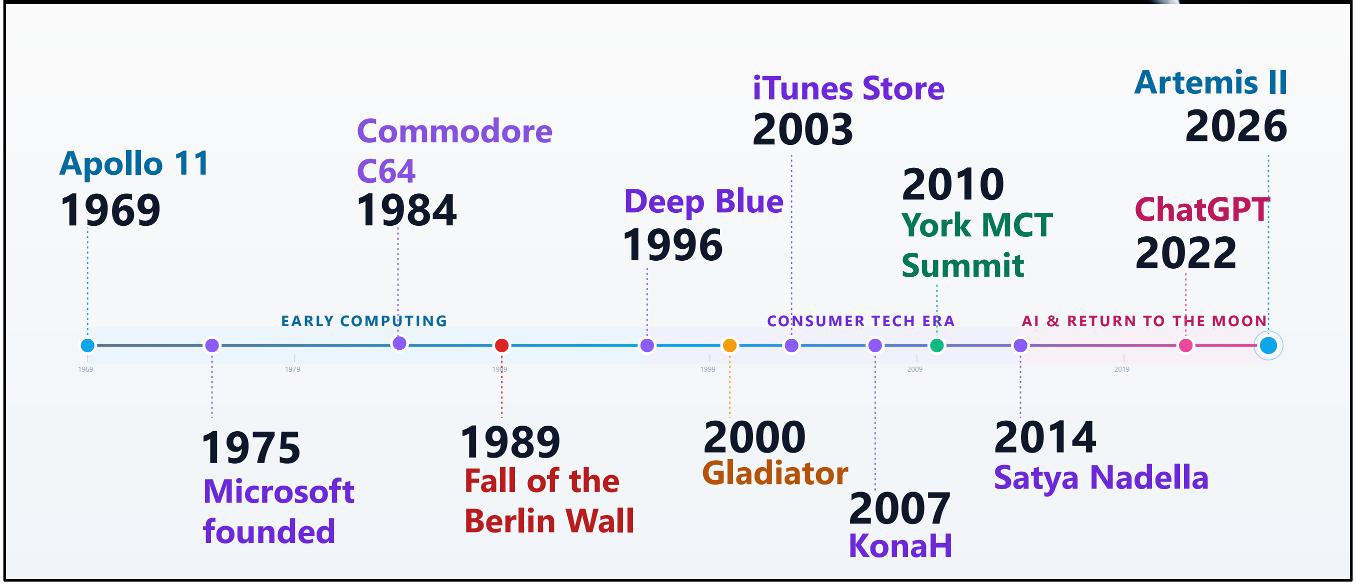


1969 — Apollo 11
On 20 July 1969, NASA astronauts Neil Armstrong and Buzz Aldrin became the first humans to walk on the Moon, while Michael Collins orbited above in the command module. The mission was the high point of the US–Soviet space race and is still one of the most-watched live events in history. Armstrong's words, "one small step for man, one giant leap for mankind," turned the landing into a shared global memory.

1975 — Microsoft is founded
Bill Gates and Paul Allen founded Microsoft (originally written "Micro-Soft") in Albuquerque, New Mexico, to sell a BASIC interpreter for the Altair 8800 home computer. At the time, software was mostly given away with hardware, and the idea of selling software as a product was new. From this small start, Microsoft grew into one of the most important companies in the history of personal computing.

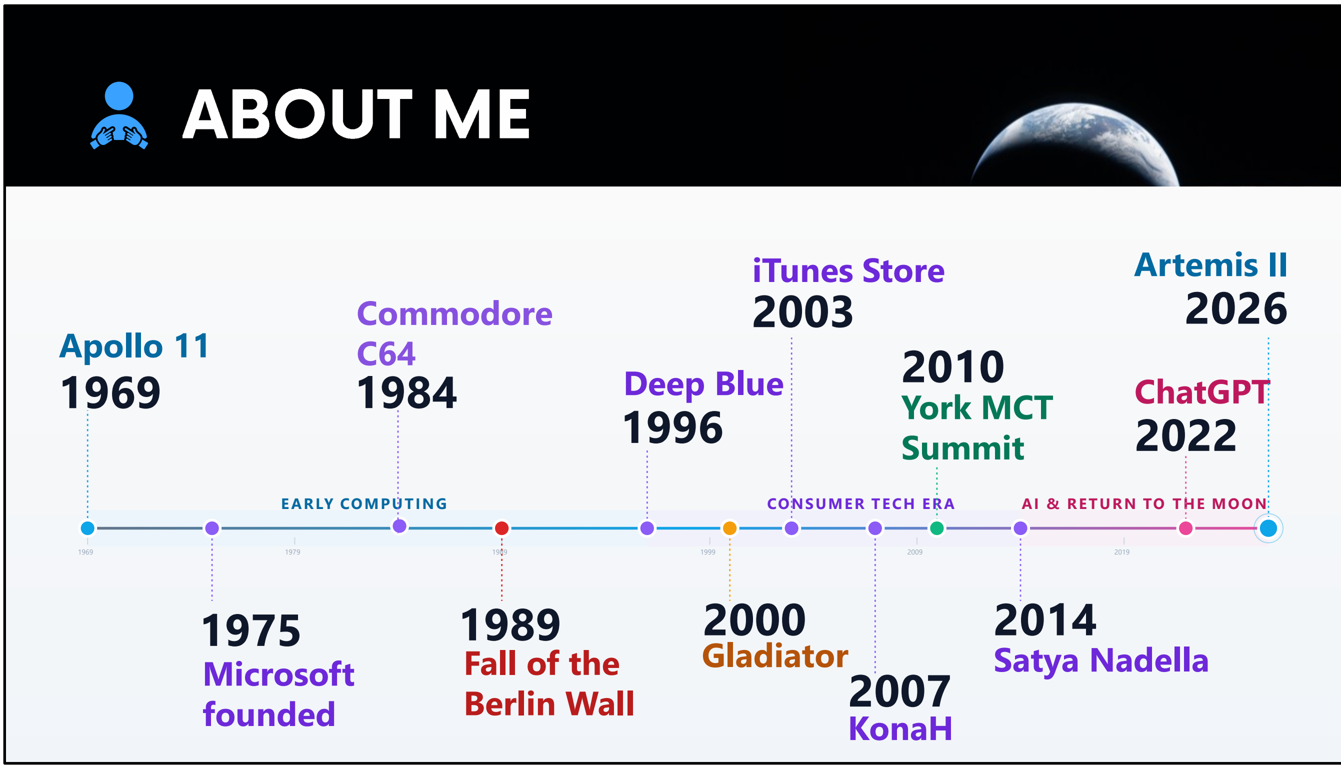
1984 — Commodore C64 shown publicly for the first time
In 1984, the **Commodore 64** was shown to the general public for the first time. Already announced in 1982, the C64 went on to become the best-selling single personal computer model of all time, with estimates ranging from 12.5 to 17 million units sold. Its combination of a low price, colour graphics, a built-in sound chip (the legendary SID), and a large software library — especially games — made it the machine that introduced an entire generation to computing. For many people in Europe, the C64 was their first real contact with a keyboard and a screen.

1989 — Fall of the Berlin Wall
On the evening of **9 November 1989**, after nearly 28 years of dividing East and West Berlin, the **Berlin Wall** was opened. Crowds from both sides climbed the wall, crossed the checkpoints, and began tearing it down — a scene broadcast live around the world. The event became the defining symbol of the end of the Cold War and led within a year to the reunification of Germany on 3 October 1990. For many people in Europe, and especially in Germany, 1989 divides life into "before" and "after."

1996 — Deep Blue vs. Garry Kasparov
In February 1996, IBM's chess computer **Deep Blue** played a six-game match against world champion **Garry Kasparov** in Philadelphia. Kasparov won the match 4–2, but Deep Blue became the first computer to beat a reigning world champion in a single game under standard tournament conditions. A year later, an upgraded Deep Blue would win the rematch — but 1996 was the moment the world first saw that a machine could seriously challenge the best human mind at chess.

2000 — Gladiator released
Ridley Scott's historical drama **Gladiator** was released in 2000, became a huge international hit, and later won five Academy Awards including Best Picture. Perhaps most notably, the film features an outstanding vocal performance by **Lisa Gerrard**. Also in 2000, **Steve Ballmer** took over as "Gladiator of Microsoft" from Bill Gates in January. Ballmer would lead the company through the Windows XP, Office, and early cloud years until 2014.

(t.b.c.)



2003 — Apple launches the iTunes Store
On 28 April 2003, Apple opened the **iTunes Music Store**, selling individual songs for 99 cents each and full albums for about $10. At a time when the music industry was struggling with file sharing on Napster and Kazaa, iTunes offered a legal, simple, one-click alternative — and it worked. The store sold one million songs in its first week, reached one billion songs by 2006, and by 2008 Apple had become the largest music retailer in the United States. The iTunes Store also set the template for every app store and digital content marketplace that followed.
2007 — Microsoft introduces "KonaH"
In 2007, Microsoft introduced a project known internally as **"KonaH"**. Given that Microsoft had already sold *Microsoft Press* three years earlier, the KonaH initiative was the logical next step in the decline of the training division.
2010 — First community-driven MCT Summit at the University of York
In 2010, the **University of York** hosted the first community-driven **MCT Summit** — a gathering of **Microsoft Certified Trainers** organised by **Andrew Bettany**. The York event marked the beginning of a series of events in which the MCT community demonstrated its independence by showing that it could run its own conferences, share teaching materials and support each other directly. This model was adopted for subsequent MCT Summit events held across Europe in later years.
2014 — Satya Nadella takes over Microsoft
In February 2014, **Satya Nadella** became the third CEO of Microsoft, following Bill Gates and Steve Ballmer. A few months later, Microsoft announced one of the largest rounds of layoffs in its history — more than 10,000 (in total about 18,000) employees lost their jobs, many of them from the former Nokia phone business. This marked the effective end of **Windows Phone** as a serious platform. Nadella then shifted Microsoft's focus toward the cloud (Azure), cross-platform software, and, later, AI.
2022 — ChatGPT is released
On 30 November 2022, **OpenAI** released **ChatGPT**, a free chat interface built on the GPT-3.5 language model. Within five days it had one million users, and within two months it had 100 million — the fastest adoption of any consumer app up to that point. ChatGPT brought large language models out of research papers and into everyday life, and started the wave of generative-AI tools we use today.
2026 — Artemis II
On **1 April 2026**, NASA launched **Artemis II** from Kennedy Space Center — the first crewed flight of the Space Launch System and Orion spacecraft, and the first time humans have travelled beyond low Earth orbit since Apollo 17 in 1972. The four-person crew (Reid Wiseman, Victor Glover, Christina Koch, and Canadian astronaut Jeremy Hansen) flew a roughly 10-day loop around the Moon, setting a new distance record for humans in space. Splashdown is scheduled for **10 April 2026** off the coast of San Diego — the same day this timeline was prepared.

Background picture from Artemis II: https://www.nasa.gov/image-detail/art002e009288/

Account types

Microsoft Account (MSA) vs Work or school account

- Work or school account
  https://myaccount.microsoft.com/
- MSA
  https://login.live.com
- Redirect (both types)
  https://account.microsoft.com
  https://login.microsoftonline.com
  https://login.microsoft.com
  ....

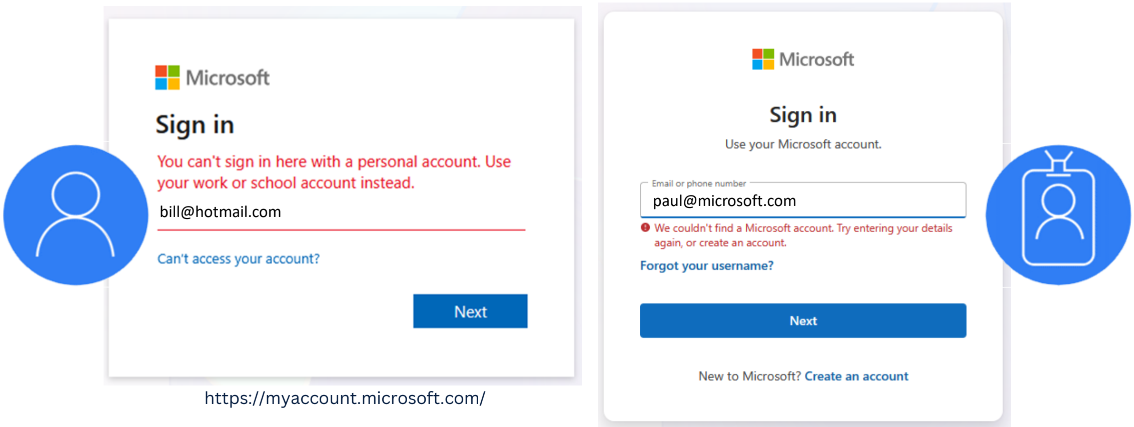
MSA: Accounts for private people / home users

- Microsoft Passport
- Microsoft Wallet
- .NET Passport
- Windows Live ID)

Domains
- hotmail.com
- outlook.com
- msn.com

References:
https://de.wikipedia.org/wiki/Datei:Zeichen_125_-_Gegenverkehr,_StVO_1992.svg
https://en.wikipedia.org/wiki/Microsoft_account

You can't sign in here



https://myaccount.microsoft.com/

https://https://login.live.com

Icon sources:
- https://learn.microsoft.com/en-us/microsoftteams/platform/toolkit/tools-prerequisites
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  https://commons.wikimedia.org/w/index.php?curid=128430642

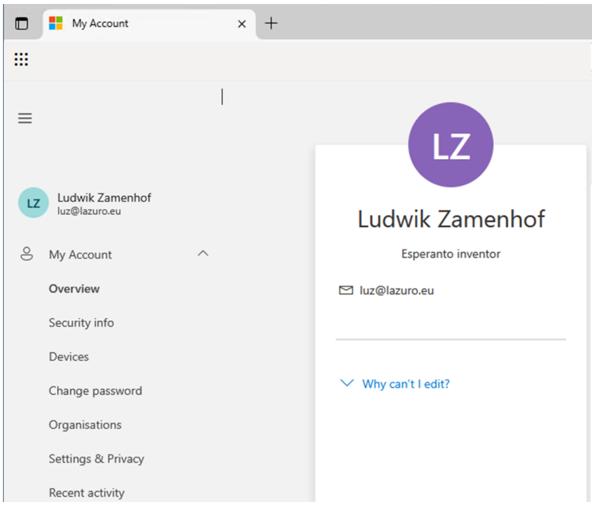
Work or school account

https://myaccount.microsoft.com/

"Microsoft work and school accounts are for organizations that use Microsoft 365 for business."

Naming is hard!

- MSOL
- Azure AD
- Entra ID



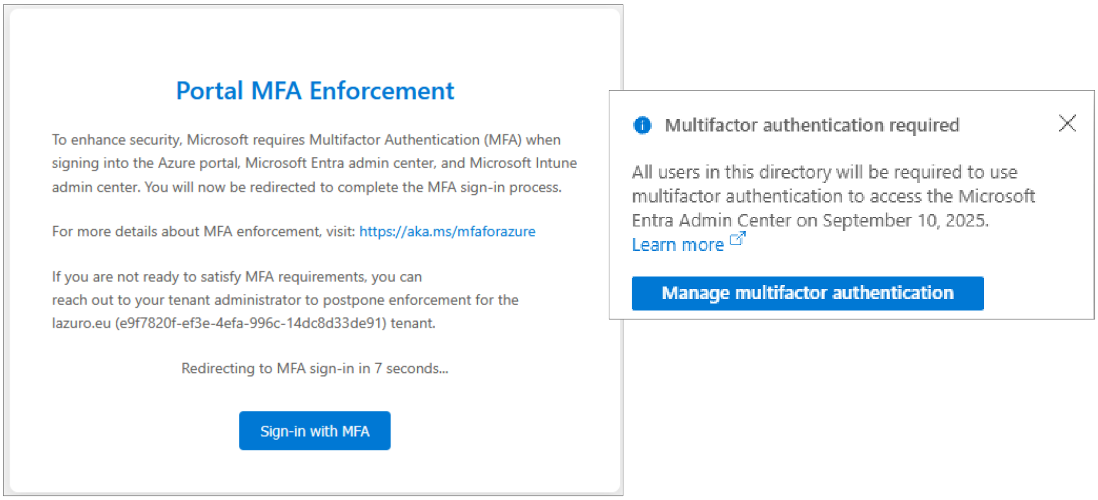
References:
https://support.microsoft.com/en-us/account-billing/what-s-the-difference-between-a-microsoft-account-and-a-work-or-school-account-72f10e1e-cab8-4950-a8da-7c45339575b0

a) LOGIN
myaccount.microsoft.com

b) Try to login: portal.azure.com
⇒ MFA enforcement

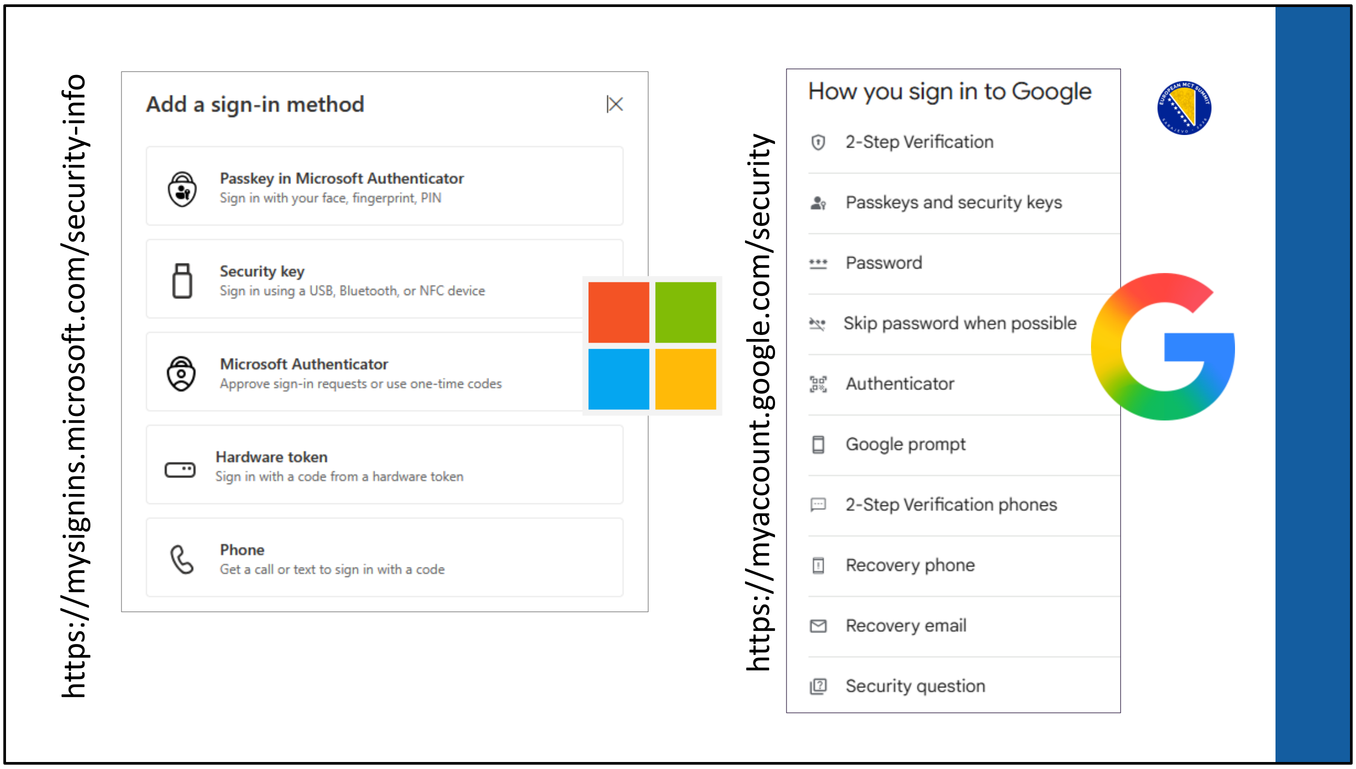
c) Back to "MyAccount"
Security Info

MFA Enforcement



Planning for mandatory multifactor authentication for Azure and other admin portals

https://learn.microsoft.com/en-us/entra/identity/authentication/concept-mandatory-multifactor-authentication?tabs=dotnet



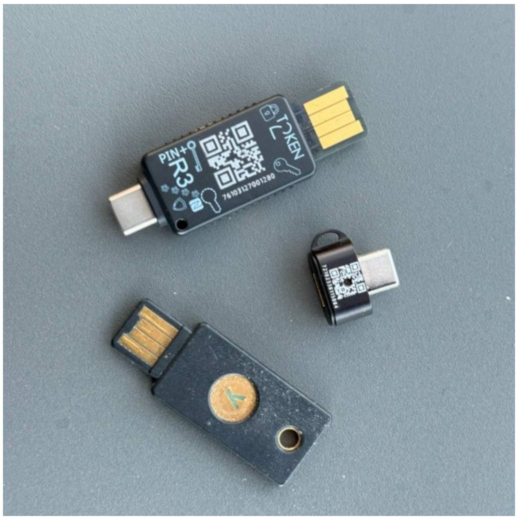
Left:

https://mysignins.microsoft.com/

Right:

https://myaccount.google.com/security

Fido2 security keys

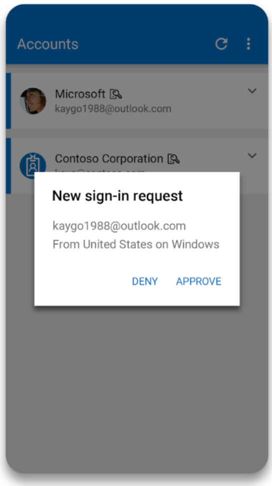


The photo shows a number of these Fido2 tokens.

Special thanks for providing the samples go to:

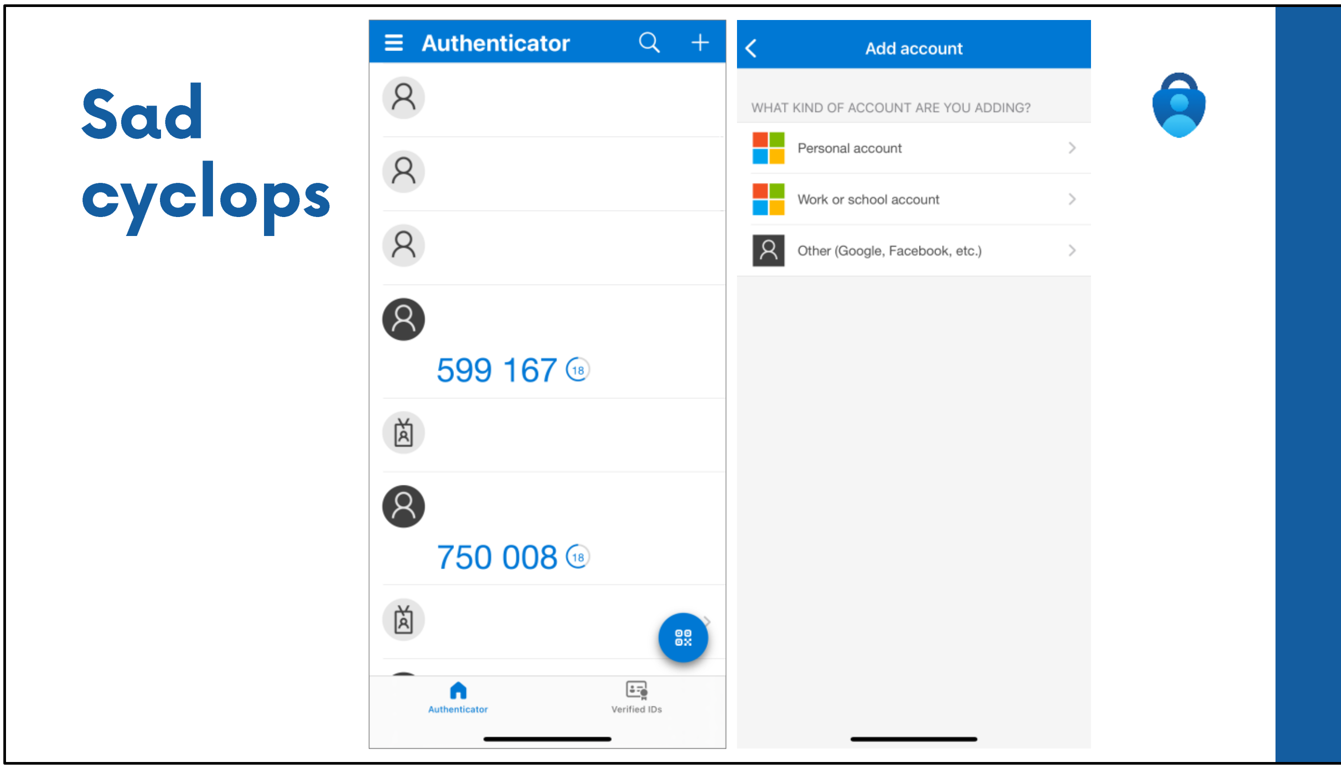
- NitroKey
- Token2
- Yubico

Microsoft Authenticator



Picture taken from Google Play Store:

https://play.google.com/store/apps/details?id=com.azure.authenticator

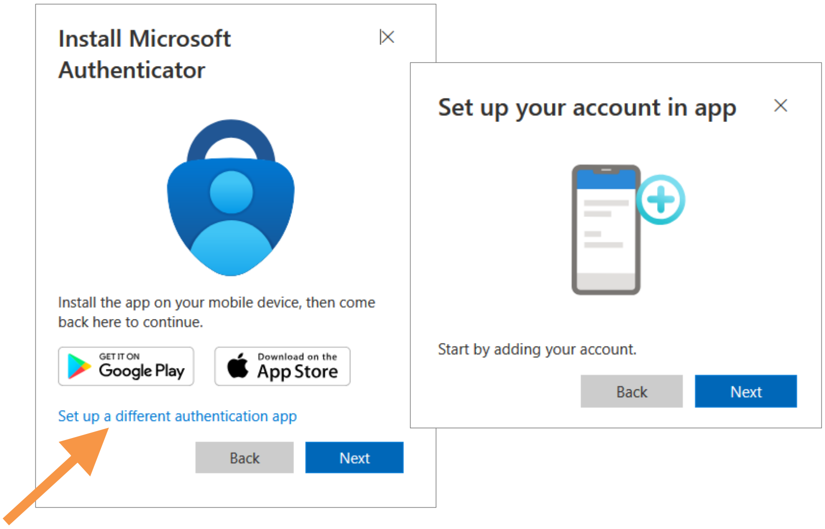


Markus Michalski coined the term “sad cyclops” to describe the various icons in Microsoft's Authenticator app: https://bsky.app/profile/markusmichalski.bsky.social/post/3lyiy6xgmzc27

You could turn this into a quiz: Which symbol represents which account type or authentication method?

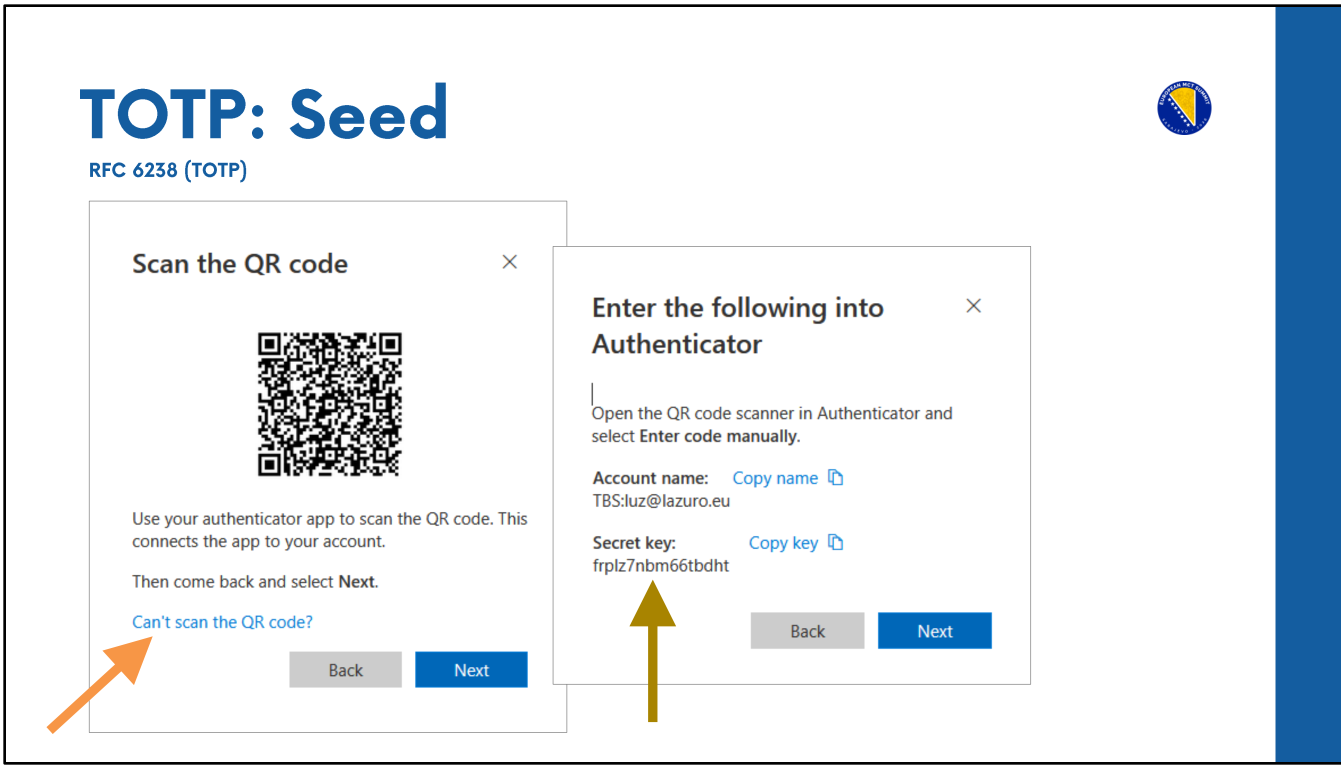
Microsoft Authenticator: TOTP

RFC 6238 (TOTP)



iOS store:
https://apps.apple.com/de/app/microsoft-authenticator/id983156458

Google play store:
https://play.google.com/store/apps/details?id=com.azure.authenticator



The shared secret, also known as SEED, can be saved and used as often as needed with (different) TOTP clients to calculate the authentication code.

The OTP module for PowerShell that I also use can be found in the PowerShell Gallery. https://www.powershellgallery.com/packages/OTP

Pictures taken from Google Play Store: https://play.google.com/store/apps/details?id=com.azure.authenticator

The (binary) seed

frplz7nbm66tbdht

00101100010111101011110011111101101000010110011110111101001100001000110011110011

The seed contains (in this case) of 16 characters representing 80 bits.

How is that possible?

The answer is: Base32 encoding

Learn more about it in the repo of the OTP module:

https://github.com/thorstenbutz/otp

BASE 32 Alphabet

RFC 4648

Char	Binary	Char	Binary	Char	Binary	Char	Binary
A	00000	I	01000	Q	10000	Y	11000
B	00001	J	01001	R	10001	Z	11001
C	00010	K	01010	S	10010	2	11010
D	00011	L	01011	T	10011	3	11011
E	00100	M	01100	U	10100	4	11100
F	00101	N	01101	V	10101	5	11101
G	00110	O	01110	W	10110	6	11110
H	00111	P	01111	X	10111	7	11111

https://www.rfc-editor.org/rfc/rfc4648#section-6

The seed: 80 bit

RFC 6238 (TOTP)

f	r	p	l	z	7	n	b
00101	10001	01111	01011	11001	11111	01101	00001

m	6	6	t	b	d	h	t
01100	11110	11110	10011	00001	00011	00111	10011

The binary seed in chunks of 5 bit encoded to 16 characters

Calculating the OTP code

RFC 6238 (TOTP)

```
# > # Generate a random seed
# > $seed = [byte[]]::new(20)
# > [Security.Cryptography.RNGCryptoServiceProvider]::new().GetBytes($seed)
# > $base32Seed = [OtpNet.Base32Encoding]::ToString($seed)
# > # Generate TOTP code
# > $totp = [OtpNet.Totp]::new([OtpNet.Base32Encoding]::ToBytes($base32Seed))
# > $code = $totp.ComputeTotp()
```

You find the complete source code including guidance in the GH repo:

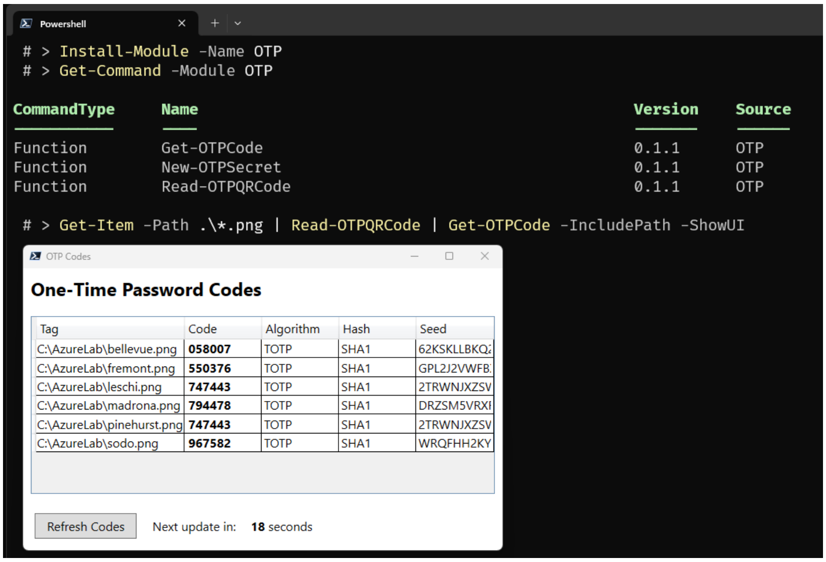
https://github.com/thorstenbutz/otp

a) LOGIN
myaccount.microsoft.com

b) Generate the OTP code

OTP module

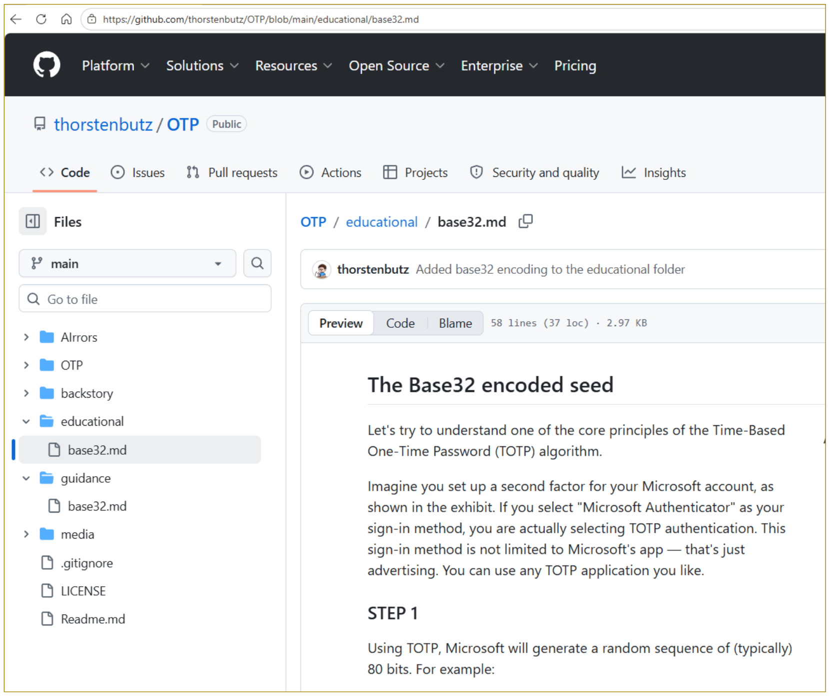
Made for MCTs



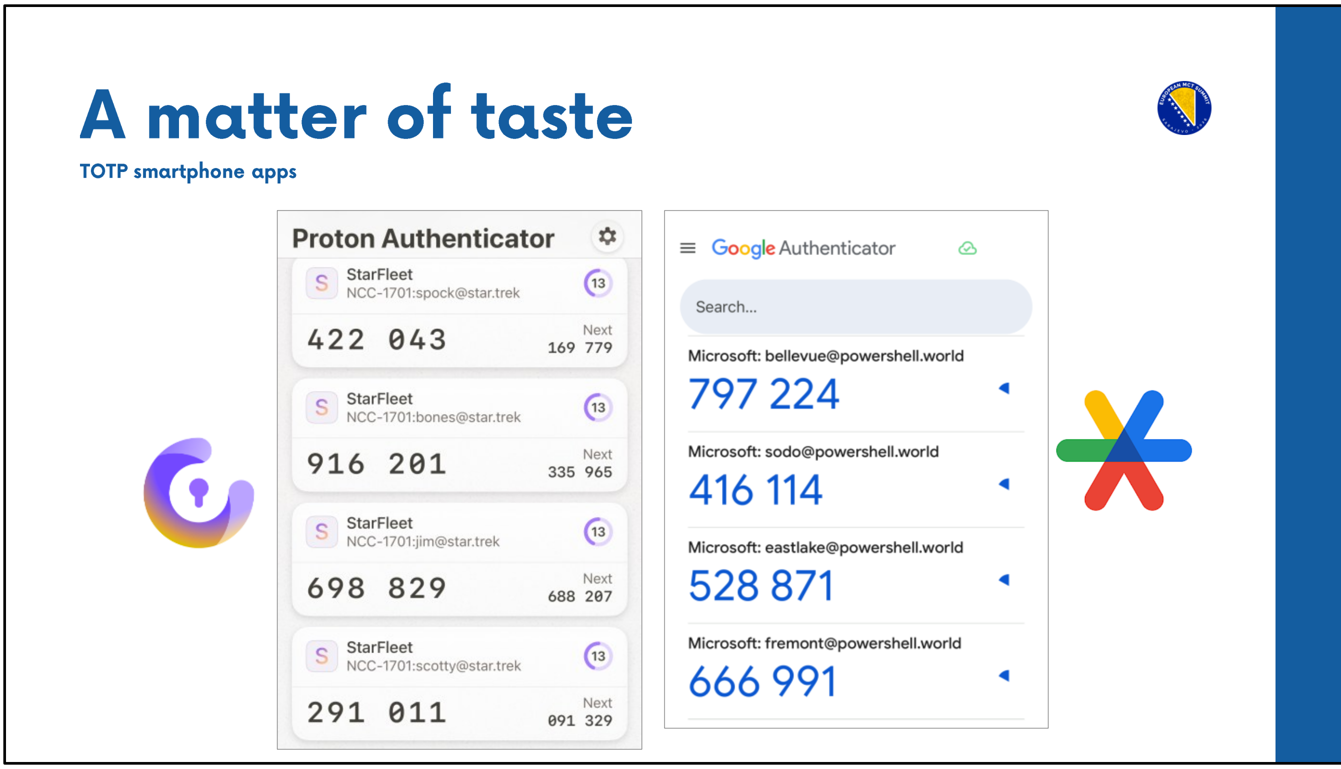
The PowerShell module is particularly well-suited for easily implementing two-factor authentication in training and test environments.

OTP

PowerShell module



Mind the educational material in the repository.



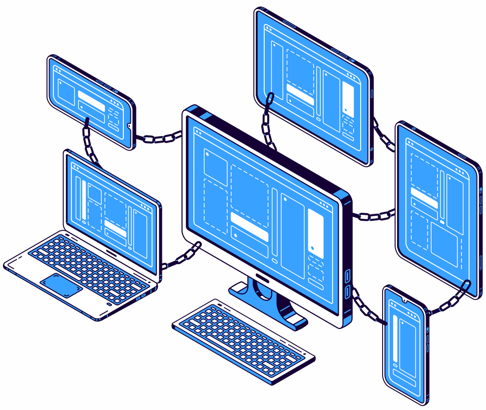
A Matter of Taste
My personal recommendation:
https://proton.me/de/authenticator

Google Authenticator holds a special place because it is by far the most widely used authenticator. If users are already using it, I see no reason to advise against it.

Wrap-up

Your mileage may vary

- A TOP client, implemented as a PowerShell module, solves many practical problems in the classroom.
- Real (Azure) lab environments enable a whole new level of teaching quality.
- Lab providers have their own solutions.



TOTP is worth a look. Despite the fact that there are "better" / more sophisticated standards such as Paskkeys (with or without the use of FIDO2 tokens), TOTP solves problems. Especially useful in lab environments.

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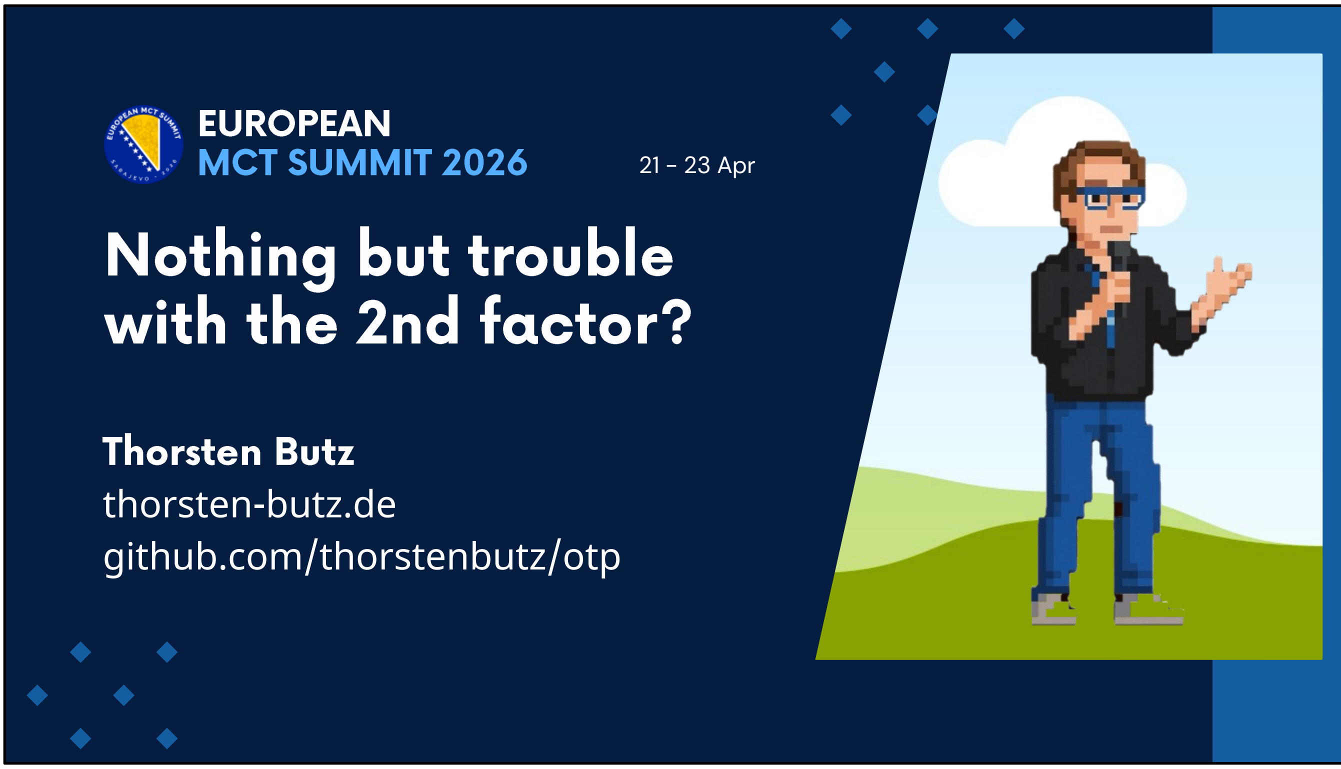
https://mctsummit.eu/2026-sarajevo-ba

Hvala Sarajevo!



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